

NE 511: Multi-Physics of Nuclear Reactors

UNIT 2: Short-Time Multi-Physics Phenomena in Reactor Core

Lecture 4 : EMISSION SPECTRA OF DELAYED NEUTRONS

THEORETICAL BACKGROUND FOR CALCULATION OF KINETICS DATA

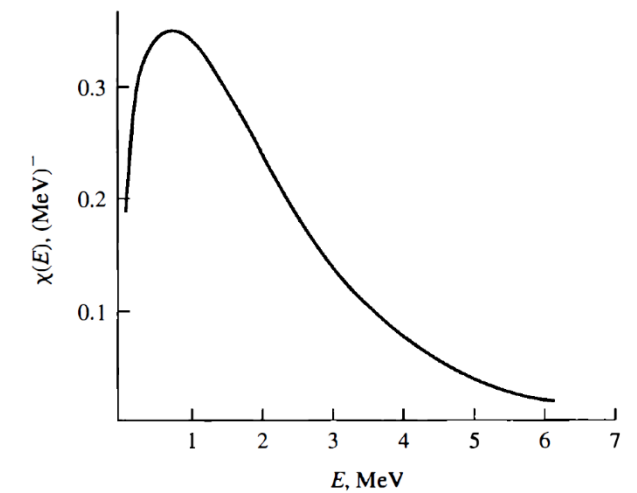
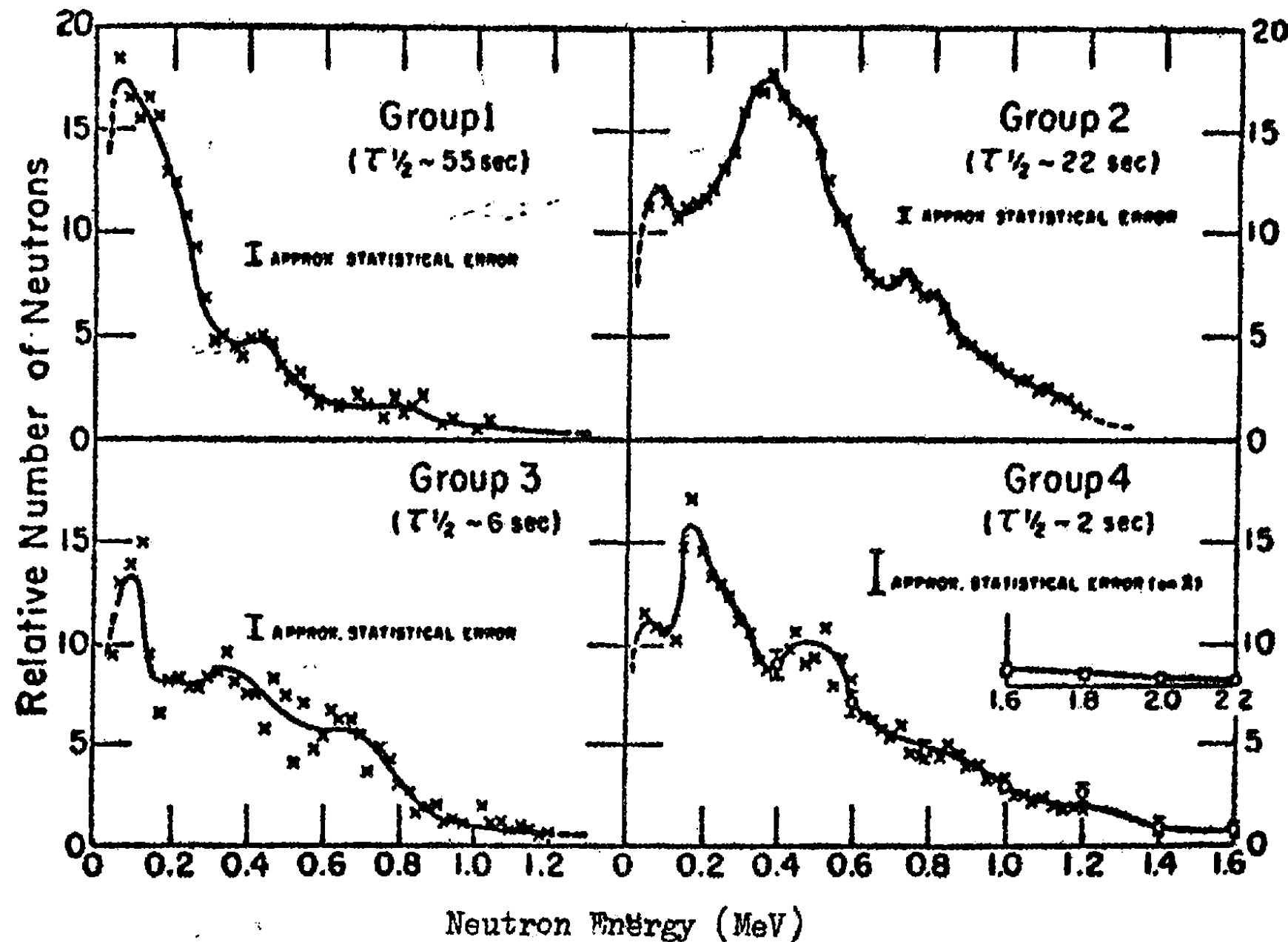
EVALUATION OF DELAYED NEUTRON DATA

Emission Spectra of Delayed Neutrons

EMISSION SPECTRA OF DELAYED NEUTRONS

- The excitation energies of delayed neutron emitters are much smaller than the excitation energies of prompt neutron emitters:
 - *the prompt fission neutrons have an average energy near 2 MeV and their spectrum extends even beyond 10 MeV;*
 - *the delayed neutrons are emitted with a much smaller average energy and thus with a softer spectrum than prompt neutrons.*

EMISSION SPECTRA OF DELAYED NEUTRONS



Prompt neutron spectrum

A. E. Evans, "Delayed Neutrons from Fission: Present Status of Measurements of Yields, Group Half-Lives and Abundances, and Spectra"

EMISSION SPECTRA OF DELAYED NEUTRONS

- Different delay groups are composed of different precursors with different energy ranges ($E^* - E_{B_n}$):
 - *the emission spectra, $\chi_{dk}(E)$, depend on the delay group*
 - *energy-dependent delayed neutron source:*

$$S_d(E, t) = n_f \sum_k \chi_{dk}(E) \nu_{dk} \lambda_k \exp(-\lambda_k t)$$

EMISSION SPECTRA OF DELAYED NEUTRONS

Considering three extreme cases:

$$S_d(E, t) = n_f \sum_k \chi_{dk}(E) \nu_{dk} \lambda_k \exp(-\lambda_k t)$$

1. The "initial" emission spectrum,

$$S_d(E, 0) \propto \sum_k \chi_{dk}(E) \nu_{dk} \lambda_k$$

2. The stationary emission spectrum, which results from integrating the energy-dependent delayed neutron source over all times,

$$S_d^{stat}(E) = \propto \sum_k \chi_{dk}(E) \nu_{dk}$$

3. The asymptotic emission spectrum (the longest living group provides the overriding contribution),

$$S_d^{as}(E) = \propto \chi_{dk}(E), \quad k = 1$$

EMISSION SPECTRA OF DELAYED NEUTRONS

- Considering these three extreme cases, the comparison shows that:
 - 1. The total delayed neutron emission spectrum may vary strongly with time.*
 - 2. The static spectrum may have a different composition compared to any time-dependent spectrum.*

Theoretical Background for Calculation of Kinetics Data

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

- Next slides are the basis for the specific terms and definitions used such as:
 - *direct (physical) beta*
 - *adjoint weighted beta effective*
 - *k-ratio beta effective*
 - *importance factor*

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Starting with the multi-group equations ...

$$\begin{aligned} \frac{1}{v_g} \frac{\partial}{\partial t} \Phi(\bar{r}, t) = & \nabla \cdot D_g(\bar{r}) \nabla \Phi_g(\bar{r}, t) - \Sigma_{ag}(\bar{r}) \Phi_g(\bar{r}, t) - \Sigma_{sg}(\bar{r}) \Phi_g(\bar{r}, t) + \sum_{g'=1}^G \Sigma_{sg'g}(\bar{r}) \Phi_{g'}(\bar{r}, t) \\ & + \chi_g^p (1 - \beta) \sum_{g'=1}^G v_{g'} \Sigma_{fg'}(\bar{r}) \Phi_{g'}(\bar{r}, t) + \chi_g^d \sum_{k=1}^N \lambda_k C_k(\bar{r}, t) + S_g^{ext} \end{aligned}$$

$$\frac{\partial}{\partial t} C_k(\bar{r}, t) = -\lambda_k C_k(\bar{r}, t) + \beta_k \sum_{g'=1}^G v_{g'} \Sigma_{fg'}(\bar{r}) \Phi_{g'}(\bar{r}, t)$$

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Note that:

- The energy spectrum of the prompt fission neutrons, is not the same as the energy spectrum of the delayed neutrons:
 - *the average prompt neutron energy is approximately 2 MeV;*
 - *the average energy of delayed neutrons is approximately 0.45 MeV.*
- In 2-group applications, all neutrons would be born in fast group:
In other words, delayed neutron effects are not taken into account.

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Beta-Effective vs. Direct Beta

Beta-Effective:

To compensate this shortcoming of 2-group applications, delayed neutron fractions are adjusted to so-called effective delayed neutron fractions, which supposedly represent the impact of the difference between delayed neutron spectra and prompt neutron spectra.

Direct Beta:

The delayed neutron fractions for a given material composition for a given energy group are generated by summation and averaging with the fission neutron production rate.

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Direct Beta

$$\beta_k \equiv \beta^{ph} = \frac{\sum_i \beta_i v_{i,k} \sum_g \sum_j v \Sigma_{f,j,i,g} V_j \Phi_{j,g}}{\sum_i \sum_g \sum_j v \Sigma_{f,j,i,g} V_j \Phi_{j,g}}$$

k - delayed neutron group; g - multi-group index;

i - fissionable nuclide index; j - region index;

β_i - total fraction of delayed neutrons from nuclide i ;

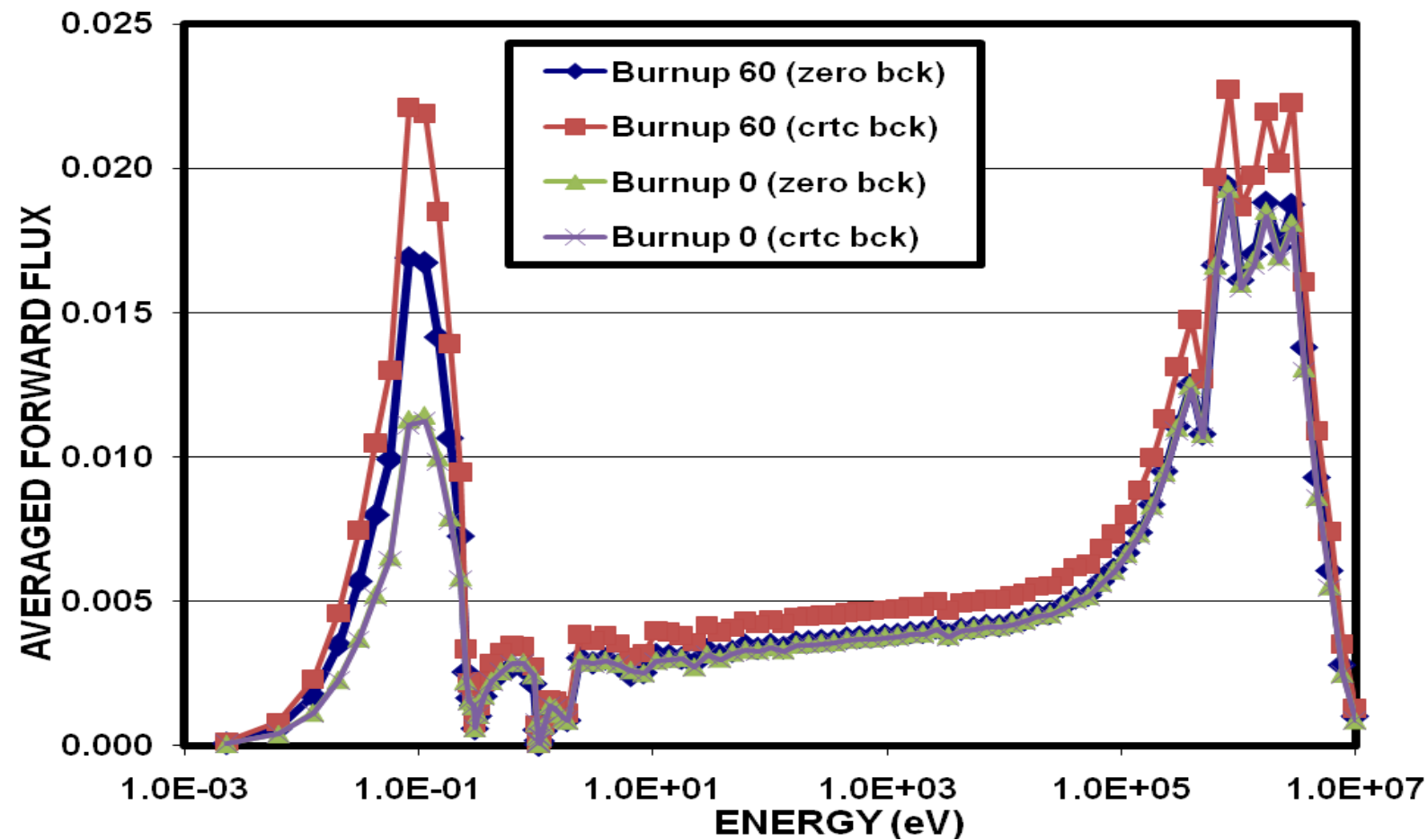
$v_{i,k}$ - the fraction of delayed neutrons from nuclide i related to delayed neutron group k

V – volume;

$\sum_j v \Sigma_{f,j,i,g} V_j \Phi_{j,g} = P_{i,g}$ - production term

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Direct Beta



The forward flux (spectrum) for a representative BWR assembly as predicted by the lattice physics code TransLAT

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Importance Factor

Importance factor (called also effectiveness factor) can be defined as a correction applied to the direct beta to compensate the inherent deficiency of the two-group model which can not explicitly capture neutron emission spectrum effects.

- The adjoint weighted (AW) importance factor:
$$I_{AW} = \frac{\sum_g \chi_g^d \Phi_g^*}{\sum_g \chi_g \Phi_g^*}$$
- The simplified k-ratio importance factor:
$$I_{k-ratio} = \left(\frac{k_d}{k} \right)_{simplified}$$

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Beta Effective

Adjoint Weighted Beta Effective

- The delayed neutron fraction can be calculated according to the adjoint spectrum, which is computed using the multi-group fundamental mode equation.
- This method provides more accurate estimate of reactivity, because it uses adjoint flux as a weighting function in the derivation of kinetics equations by eliminating first-order variations in the flux spectrum.

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Beta Effective

Adjoint Weighted Beta Effective

Adjoint weighted delayed neutron fraction can be formulated as:

$$\beta_d^{eff} = \frac{\sum_i \beta_{i,k} \sum_g \nu \Sigma_{fg,i} \Phi_g (\sum_{g'} \chi_{k,g'}^d \Phi_{g'}^*)}{\sum_i \sum_g \nu \Sigma_{fg,i} \Phi_g \sum_{g'} \Phi_{g'}^* (\chi_{g'}^p (1 - \beta_i) + \sum_k \chi_{k,g'}^d \beta_{i,k})}$$

where $\beta_i = \sum_{k=1}^6 \beta_{i,k}$

and the adjoint weighted importance factor can be defined as: $I = \frac{\beta_k^{eff}}{\beta_k}$

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Beta Effective

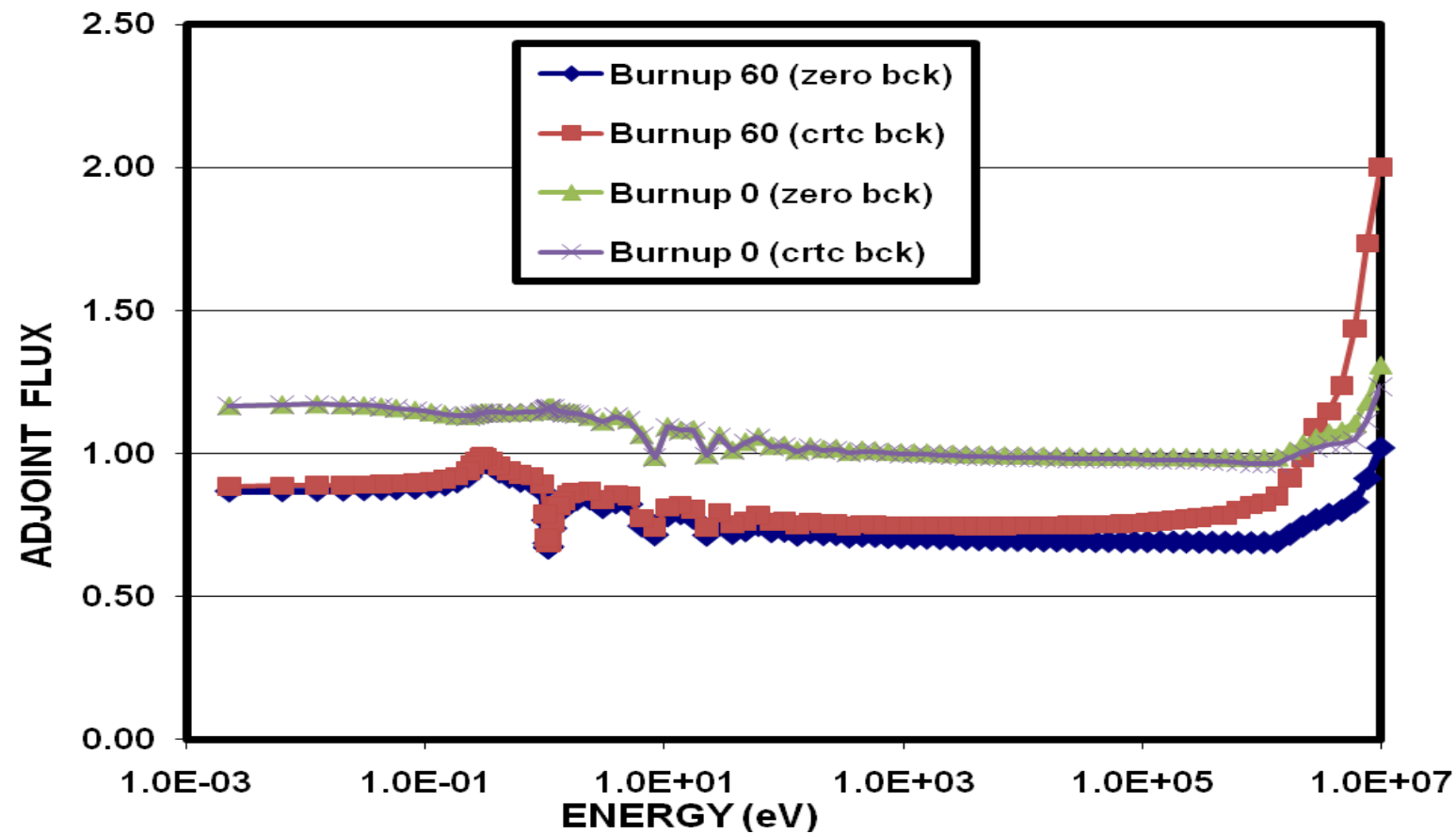
Adjoint Weighted Beta Effective

- The real and adjoint spectra are used to calculate the effective delayed neutron fraction.
- Since the energy of the delayed fission spectrum is lower than that of the prompt fission spectrum, delayed neutrons have a smaller chance to leak out of the system.
- This is taken into consideration in the kinetics equations by weighting the physical beta with the energy dependent adjoint spectra which quantifies the energy importance of neutrons in the system.

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Effective Beta

Adjoint Weighted Beta Effective



The adjoint flux (spectrum) for a representative BWR assembly as predicted by the lattice physics code TransLAT

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Beta Effective

k-ratio Beta Effective

- This method also takes into account the fact that delayed neutrons have smaller chance to leak out of the system since average energy of delayed neutron spectrum is lower than the prompt neutron spectrum.
- In other words, “k-ratio Beta Effective (β^{eff})” will refer to the physical beta adjusted by means of the importance factor (I) determined by k-ratio method:

$$I = \frac{k_d}{k} \quad \text{and} \quad \beta_k^{\text{eff}} = I\beta_k$$

k is the static multiplication factor.

k_d is the so-called delayed neutron multiplication factor.

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Beta Effective

k-ratio Beta Effective

- In *k*-ratio method, the average delayed neutron source can be used at 0.45 MeV and k_d should be calculated separately:

$$k_d = \frac{P - \sum_{h=1}^{h_d} P_h}{Abs - \sum_{h=1}^{h_d} Abs_h + Leak - \sum_{h=1}^{h_d} Leak_h}$$

- Production (P), absorption (Abs) and leakage (Leak) terms are given by the cell neutron balance in the macro-group (h)
- h and h_d can be defined by: $E_{h_d+1} \geq 0.45 \times 10^6 \geq E_{h_d+3}$

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Beta Effective

k-ratio Beta Effective

The definition of the k-ratio method can be stated as:

$$\frac{k_d}{k} = \frac{\int_0^{E_d} dE \chi^d(E) \int_0^{E_d} dE' \nu \Sigma_f(E') \Phi(E')}{\int_0^{E_d} dE (\Sigma_a(E) + D(E) B^2) \Phi(E)} \frac{\int_0^{\infty} dE (\Sigma_a(E) + D(E) B^2) \Phi(E)}{\int_0^{\infty} dE \chi(E) \int_0^{\infty} dE' \nu \Sigma_f(E') \Phi(E')}$$

where B^2 is a buckling representing leakage effects, $\chi^d(E)$ and $\chi(E)$ are the delayed neutron and the total neutron emission spectra, and E_d represents the upper energy boundary of the delayed neutron emission spectrum.

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Beta Effective

k-ratio Beta Effective

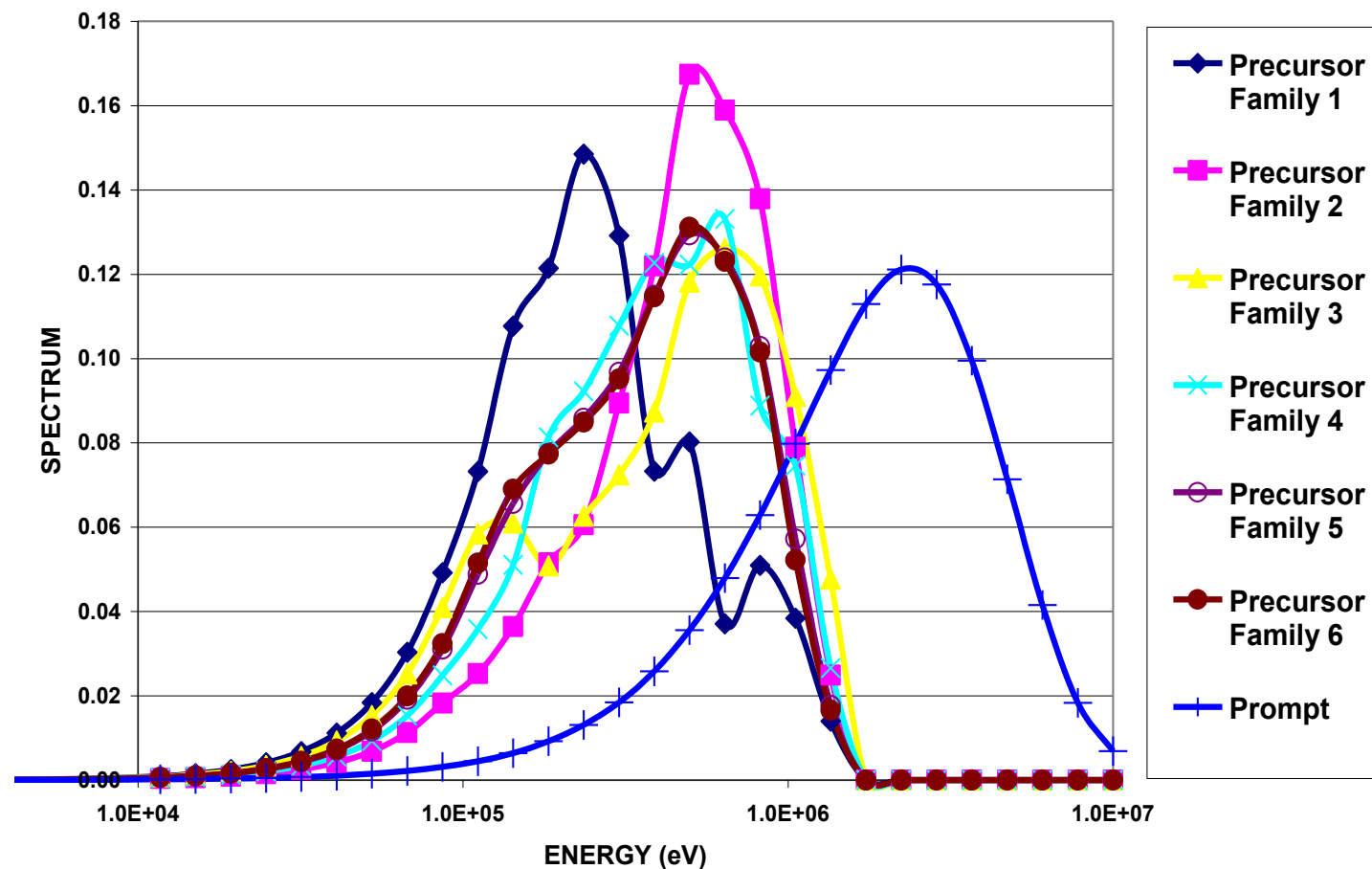
- The fission spectra are normalized to unity since they represent probability density distributions, and their integral values can therefore be dropped from the above expression.
- Although the flux spectra obtained from the two eigenvalue calculations are not the same, it is assumed that they are the same, so the simplified k-ratio method is obtained by the following equation:

$$\left(\frac{k_d}{k}\right)_{simplified} = \frac{\int_0^E dE' \nu \Sigma_f(E') \Phi(E')}{\int_0^E dE (\Sigma_a(E) + D(E) B^2) \Phi(E)} \frac{\int_0^\infty dE (\Sigma_a(E) + D(E) B^2) \Phi(E)}{\int_0^\infty dE' \nu \Sigma_f(E') \Phi(E')}$$

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Beta Effective

k-ratio Beta Effective



97-Group fission spectra for each delayed neutron precursor family and prompt fission spectrum for unrodded BWR assembly as calculated by TransLAT

THEORETICAL BACKGROUND FOR KINETICS CALCULATION

Thus:

“Direct Beta β ” will refer to the fission rate averaged beta.

“k-ratio Beta Effective β^{eff} ” will refer to the direct beta adjusted by means of the importance factor (I) determined by k-ratio method.

“Adjoint Weighted Beta Effective β^{eff} ” will refer to the effective beta computed using adjoint weighting.

Evaluation of Delayed Neutron Data

EVALUATION OF DELAYED NEUTRON DATA

NEA WPEC SG6

- Delayed neutron data uncertainties may result in undesirable conservatism in the design and operation of nuclear power plants.
- An international working group was formed at the Nuclear Energy Agency (NEA), OECD - NEA WPEC SG6 (Working Party on International Nuclear Data Evaluation Co-operation, Sub-Group 6) - to review, improve and validate delayed neutron (DN) data.
- The SG6 has reviewed the DN data at three different levels:
 - *individual precursor (microscopic) level;*
 - *aggregate precursors (macroscopic) level;*
 - *in-pile (integral) measurement level.*

EVALUATION OF DELAYED NEUTRON DATA

NEA WPEC SG6

- NEA WPEC SG6 group completed its work with the following major results:
 1. *Recommending a new set of DN yields for the major fissile isotopes.*
 2. *Developing a new 8-group model that has a better physical basis than the presently used 6-group model.*
 3. *Investigating the need that the DN group parameters be made an explicit function of the incident neutron energy.*

EVALUATION OF DELAYED NEUTRON DATA

NEA WPEC SG6

- In summary, the studies performed for the delayed neutron data have addressed the following issues:
 - *the beta-effective measurements;*
 - *the energy dependence of delayed neutron yields;*
 - *evaluations of total delayed neutron yields;*
 - *time-dependent data;*
 - *the energy spectra of delayed neutrons.*

EVALUATION OF DELAYED NEUTRON DATA

Beta-effective Measurements (1/2)

- Measurements at microscopic level and the associated evaluations are being incorporated in the fission product yield and radioactive decay data files in the major nuclear data libraries such as ENDF/B, JEF, and JENDL.
- Although the accuracy of the data on microscopic level has improved (for β_{eff} calculations) the reliance must be still placed on the macroscopic measurements of the DN emission data for major isotopes and validation of the data using integral reactor measurements of β_{eff} and time-dependent effects.

EVALUATION OF DELAYED NEUTRON DATA

Beta-effective Measurements (2/2)

- There is a tendency for the measurements made in the different experimental programs to yield different values of β_{eff} .
- *The uncertainties associated with measurements are estimated and a use of an average of the adjusted values based on the different studies is suggested.*
- *It is considered that using these averaged values, an accuracy of $\pm 3\%$ will be achieved in β_{eff} calculations for conventional **thermal reactors** and $\pm 2\%$ to $\pm 3\%$ for **fast reactors** fuelled with the major actinide isotopes.*

EVALUATION OF DELAYED NEUTRON DATA

Energy Dependence of Delayed Neutron Yields (1/2)

The incident neutron energy dependence of total delayed neutron yields is believed to be small - the difference between thermal reactor and fast reactor spectrum averaged values is at most a few percent.

- Fission product precursor summation calculations give much larger differences and the reason for these needs to be understood.
- The uncertainty about the possible variations through resonances also prevents a clear conclusion being reached about the relationship between thermal and fast spectrum yields.

EVALUATION OF DELAYED NEUTRON DATA

Energy Dependence of Delayed Neutron Yields (2/2)

- Very accurate relative measurements (+1%) will be required for the major actinide isotopes.
- For the secondary isotopes, the uncertainties are larger and more measurements having a better precision would be useful.
- Accurate relative measurements of the energy dependence of total yields, and of the fractional yields used to represent time dependence, would enable the systematic of the interrelation-ships to be explored in more detail.
- SG6 has recommended the following total yield values:

<u>U²³⁵ thermal</u>	<u>U²³⁵ fast</u>	<u>U²³⁸ fast</u>	<u>Pu²³⁹ thermal</u>	<u>Pu²³⁹ fast</u>
<u>0.0162</u>	<u>0.0163</u>	<u>0.0465</u>	<u>0.00650</u>	<u>0.00651</u>

EVALUATION OF DELAYED NEUTRON DATA

Energy Spectra of Delayed Neutrons (1/1)

- Improved representations of the time dependence of the delayed neutron emission was proposed based on a new 8-group precursor structure for delayed neutron group constants and spectra.
- In the new structure, the half-lives of the first three groups have been fixed at the half-lives of the three longest-lived dominant precursors.
- The new 8-group structure is characterized by the same set of half-lives for all fission isotopes and for fission induced by neutrons of different energies – set of nine differential equations (eight for the precursors in different groups and one for the neutron density).

Next Class

Preliminary Formulation of the Point Kinetics:

Prompt Neutron Balance Equation

Intuitive Point Kinetics Equation

One-Group Point Kinetics Equation